



REVACHOL / PORTFOLIO / WORK PAGE

VISUAL PASS01

Diagnosis and remediation of the Work page against the supplied maximalist glitch / neon-HUD reference set.

10 FILES CHANGED

5 FINDINGS

TSC CLEAN

DATE	27 AUGUST 2026
PROJECT	D:\WORK\PROJECT\JOB\AI\WEBSITE_AI\SITE
SCOPE	works.html - floor, atmosphere, poster treatment, overlay
STACK	Vite 7 · TypeScript · Pixi.js 8 · GSAP · pixi-filters
DIRECTION	Full maximalist (confirmed, supersedes "minimal / elegant")



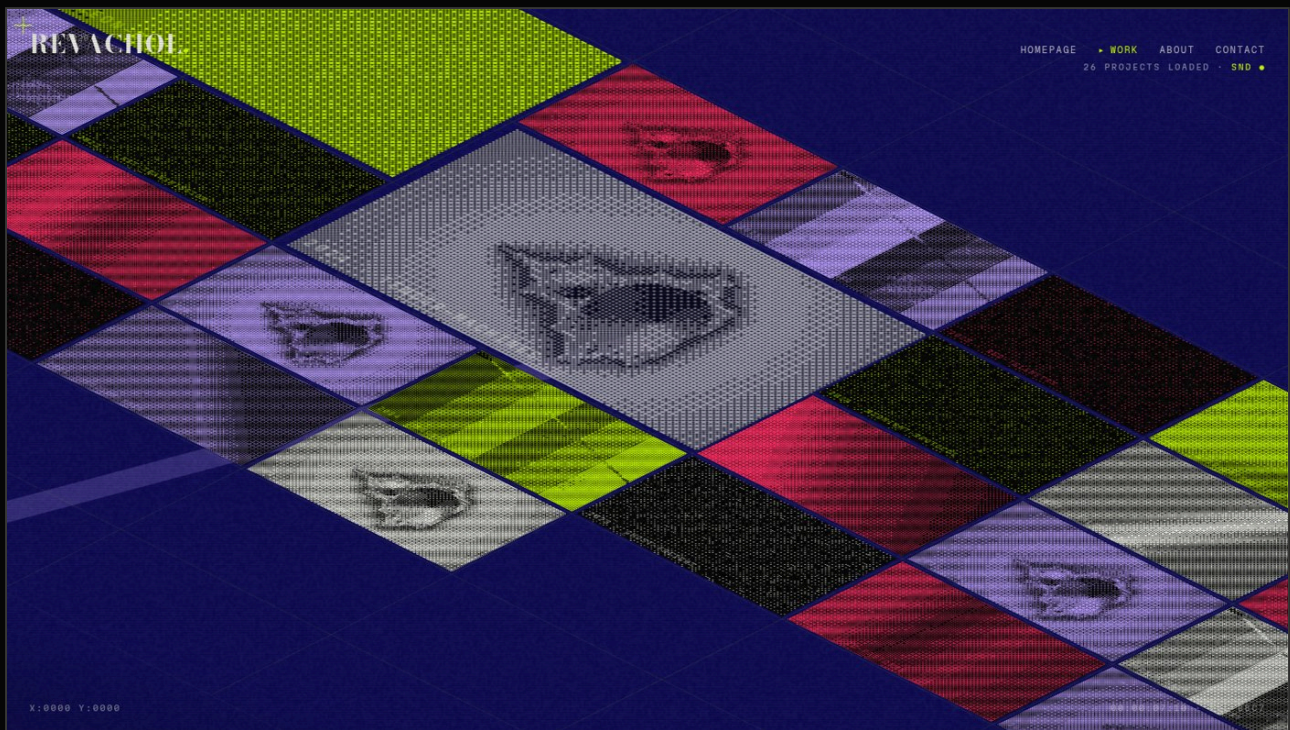
01 THE BRIEF

You supplied five reference images – an acid-green wireframe anatomy HUD, a violet RGB-split *GLITCH* poster, a datamoshed Russian theatre bill, and two flat neon vector compositions – and said the Work page looked "kinda off" against them. You also called the grain over the black background annoying.

Before touching anything I asked which direction won, since the saved direction for this site was minimal and elegant and the references are the opposite. You chose **full maximalist**. Everything below follows from that.

HOW THE PAGE WAS INSPECTED

The Chrome extension was not connected, so `localhost:5173` was unreachable from this session. Instead the source, fonts, content manifests and all 26 posters were staged into this environment, `npm install` ed, and served through Vite behind headless Chromium with SwiftShader – so every screenshot in this report is the real page, WebGL and all, not a mockup.



BEFORE / works.html at 1600x900

02 DIAGNOSIS

F-01 There was no black

CRITICAL

`fields.ts` drew an ultramarine disc at radius 1250 with 32% alpha, plus a 64px-wide ring at 55%. Pushed through the 2:1 floor matrix that geometry covers the entire viewport at any realistic screen size. Your references are 60-80% black with hard-edged colour sitting *on* it. With the void painted over, acid green read olive and magenta read brick – the palette had nowhere to sit.

F-02 Everything was dithered, so nothing was graphic

CRITICAL

All 26 posters went through 1-bit Bayer duotone at 240px wide, then upsampled to 320px (640px for the 2x2 tiles). Two problems compounded: the pattern became a visible mosaic rather than a fine screen, and applying it uniformly to every tile turned the floor into noise fabric. The references use halftone as *one* texture against flat vector shapes – never as the entire surface.

F-03 No typography, no technical debris

HIGH

References 4 and 5 are roughly 80% typography and junk data – pills, chevrons, checkerboards, registration marks, code strings, ruler ticks. The Work page had four nav links and a coordinate readout. Nothing carried the poster language.

F-04 One axis only

HIGH

Tiles, debris grid and colour fields all shared the same 45° shear, so no element disagreed with any other. That reads as an Escher carpet rather than a composition. The references consistently fight an orthogonal grid against diagonal elements.

F-05 The grain was not grain

HIGH

`grain.ts` generated a 640x360 `ImageData` buffer and CSS-stretched it to `100vw × 100vh`. On a 2560px display that is a 4x bilinear upscale, so the output was soft blotches, not film grain – and it regenerated every 120ms, so the blotches crawled. At 4% opacity over `#060606` that crawl was the most visible thing on screen in the quiet areas. On top of it sat a scanline layer at `0.5` opacity and a vignette, neither of which appears anywhere in the references.

03 WHAT CHANGED

ATMOSPHERE – GRAIN.TS, BASE.CSS

Grain is now a 96px noise tile rendered at 1:1 device pixels with `image-rendering: pixelated`, so it never scales. Four frames are pre-baked once and swapped at 90ms with a randomised background offset – a tick now costs one background swap instead of 230,400 `Math.random()` calls.

The important part is the compositing. Texel values are centred on 128 and the layer uses `mix-blend-mode: overlay`. Overlay against a base darker than 50% evaluates to $2 \times \text{base} \times \text{blend}$, so against `#060606` the result is *mathematically* black regardless of the noise value. The grain now exists only where there is something to texture – posters and colour fields – and is absent on the void. That is the fix for your specific complaint, not a reduction in opacity.

Scanlines dropped from 0.5 to 0.22 opacity on a 4px pitch. The vignette layer was deleted outright, both the DOM node and the CSS rule.

GROUND – FIELDS.TS

Rewritten from a translucent wash to hard-edged furniture, sized in floor units derived from the actual carpet bounds so it scales with the project count: one flat ultramarine slab at full alpha with bone and acid edge rules, two hairline HUD rings, a heavy magenta arc, a violet echo arc, an eleven-step chevron rail, an eighteen-column checkerboard strip, four registration crosses and a forty-tick ruler rail. It rings the carpet rather than covering it.

`world.ts` now passes carpet bounds instead of a centre point, and computes a pan halo from the carpet size so the furniture is reachable – but only just, so the page never strands you in empty black.

POSTERS – DITHER.TS, WORLD.TS

Dither resolution raised from 240px to 512px, which is a 320pt card at 2× DPR, so the Bayer pattern stays a fine screen. A `contrast(1.3) saturate(1.15)` crush is applied before thresholding, because flat graded footage otherwise dithers to mud.

`ditherImageToCanvas` gained an optional `mix` parameter that composites the original back in `overlay`. A per-slug hash assigns each project a value from `[0, 0, 0.5, 0.72]`, so half the floor stays hard 1-bit graphic and half keeps its image. Deterministic, so the layout never shuffles between loads. The parameter defaults to 0, so the existing `dither.test.ts` assertions are untouched.

PANELS – TILE.TS, DEBRIS.TS

Each tile gained an accent hairline at 42% alpha, four 26px corner brackets at full alpha, a 20px label plate with a solid accent spine, an index tab, and a second mono code block derived from its lattice position. The slug label went from 9px at 55% alpha to 10px at 92%. The debris grid was lifted from 4.5% alpha, where it was invisible, to 11%.

SECOND AXIS – NEW OVERLAY.TS, COMPONENTS.CSS

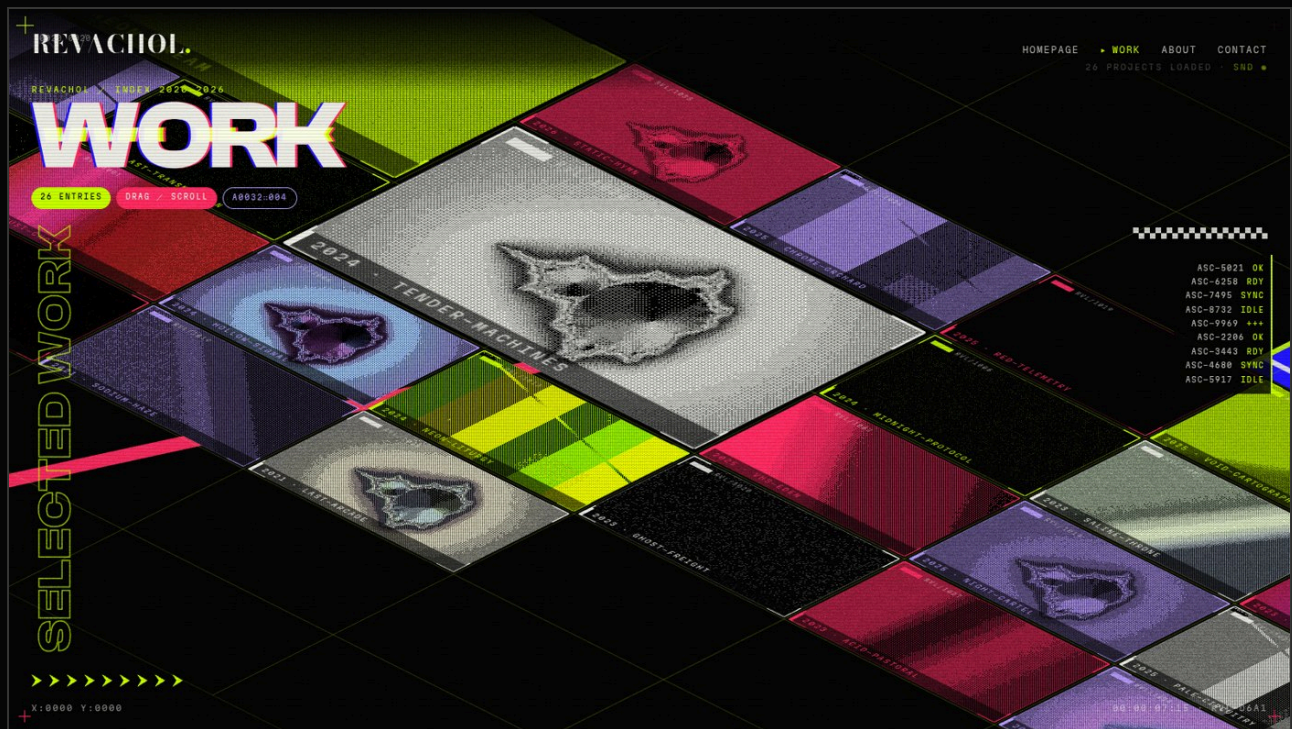
A screen-fixed, entirely inert DOM layer deliberately at odds with the shear: the *WORK* masthead in Clash Display with a two-colour RGB split and an acid slice clipped through the middle; three status pills; an outlined *SELECTED WORK* spine running up the left edge; a nine-row telemetry column on its own plate; a chevron rail; a checkerboard strip; and four corner registration marks. The Work page also gained a gradient scrim under the nav, because the brand was previously sitting directly on a lit tile.

Hover focus was softened – desaturation from -0.55 to -0.35 , dimming from 0.55 to 0.62 – so the surrounding floor keeps some colour instead of going fully grey.

FILE MANIFEST

File	Change
src/shell/grain.ts	Rewritten – tiled 1:1 noise, four pre-baked frames, overlay compositing; vignette node removed
src/styles/base.css	Atmosphere block replaced – grain as repeating background, scanlines $0.5 \rightarrow 0.22$, vignette rule deleted, reduced-motion clause
src/styles/components.css	Work overlay system added (~70 rules); nav scrim on .page-work
src/works/fields.ts	Rewritten – hard-edged floor furniture sized from carpet bounds; new CarpetBounds signature
src/works/debris.ts	Grid stroke alpha $0.045 \rightarrow 0.11$
src/works/tile.ts	Panel frame, corner brackets, label plate, index tab, secondary code block; label 9pt/55% $\rightarrow$ 10pt/92%
src/works/world.ts	Carpet bounds passed to fields; pan halo computed from carpet size; posterMix() added; dither $240 \rightarrow 512$; hover desat $-0.55 \rightarrow -0.35$
src/works/overlay.ts	New – the orthogonal poster layer
src/lib/dither.ts	Optional mix parameter; pre-threshold contrast crush. Backwards-compatible
src/pages/works.ts	Overlay mounted after the project count is known

04 RESULT



AFTER / default view

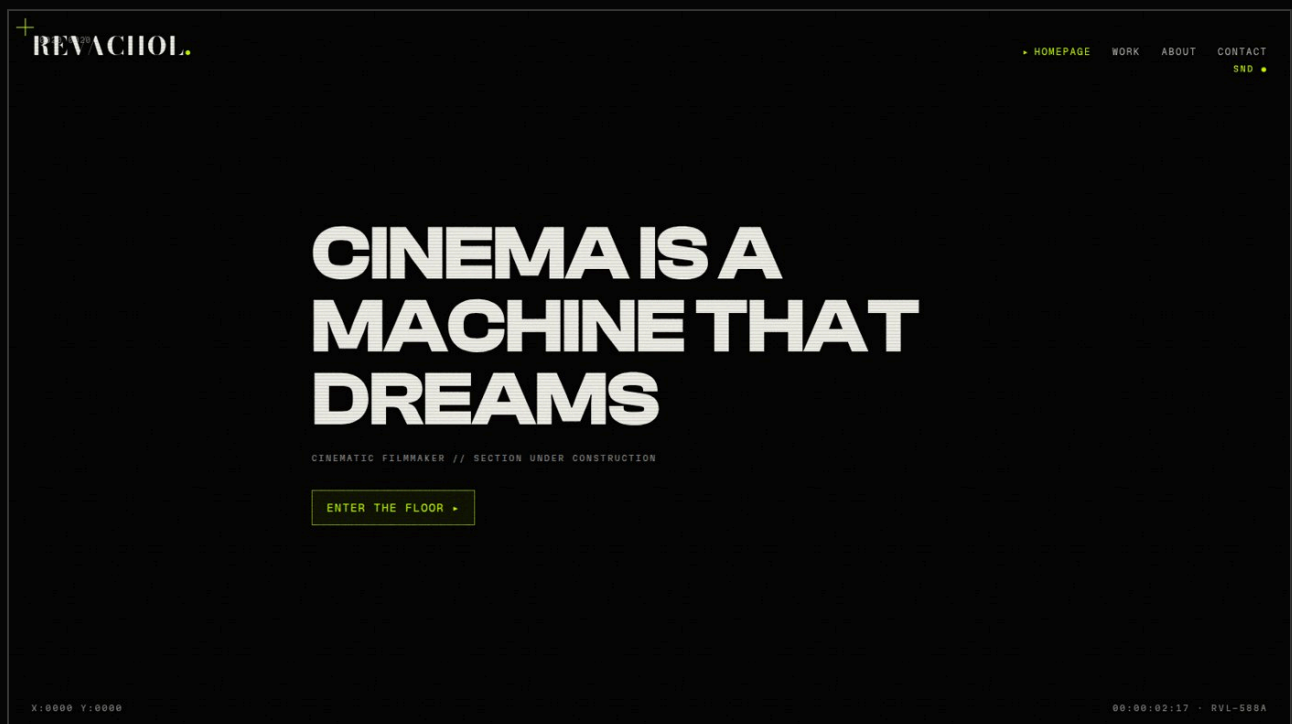


AFTER / hover - tile lifts upright and magnifies, floor desaturates, Bodoni title scrambles in

05 VERIFICATION

Check	Result
TypeScript – <code>tsc --noEmit</code> , run on your machine after the write	PASS
Rendered screenshots, headless Chromium + SwiftShader, 1600×900	PASS
Hover / focus interaction path	PASS
Homepage regression (atmosphere layers are global)	PASS
Console – no errors beyond expected 404s for absent preview videos	PASS
Unit tests – <code>vitest</code>	BLOCKED
Real device / DPR-2 display, mobile breakpoint, reduced-motion	NOT RUN

The `vitest` block is pre-existing and unrelated: your `node_modules` resolves a rollup native binary that the runner could not load (`MODULE_NOT_FOUND` in `rollup/dist/native.js`). No test file was modified and the only touched function signature is backwards-compatible, so the suite should pass once that install is repaired.



REGRESSION / homepage with the new atmosphere – void is clean, texture survives only on the type

R-01

The content is still placeholder

All 26 projects are ffmpeg synthetic sources – `gradients`, `mandelbrot`, `testsrc2` and `life` – generated by `scripts/gen-placeholders.ps1`. A meaningful share of any remaining muddiness is the placeholder, not the design. Judge the poster treatment properly only once real stills are in. The dither crush was tuned for graded footage, not for test patterns.

R-02

Scanline weight is a taste dial

At 0.22 opacity on a 4px pitch the scanlines are clearly visible across large display type on the homepage. Reference 2 supports that; references 4 and 5 do not. One number in `base.css` if you want it lighter.

R-03

Verify on your own display

Everything here was shot at 1600×900 and DPR 1. The grain fix specifically depends on device-pixel behaviour, so it is worth a look at DPR 2 – the tile is set in CSS pixels and may want `image-rendering` tuning on a HiDPI panel.

R-04

The overlay is desktop-tuned

Below 900px the spine, telemetry and checkerboard are hidden and the masthead reflows. That is a reasonable default but it has not been eyeballed on a real handset.

R-05

Remaining pages are untouched

About, Contact and the project detail template still carry the earlier, quieter language. If the maximalist direction is settled, they will need the same pass – the tokens and atmosphere are already shared, so the work is the overlay layer and the per-page composition rather than the system.